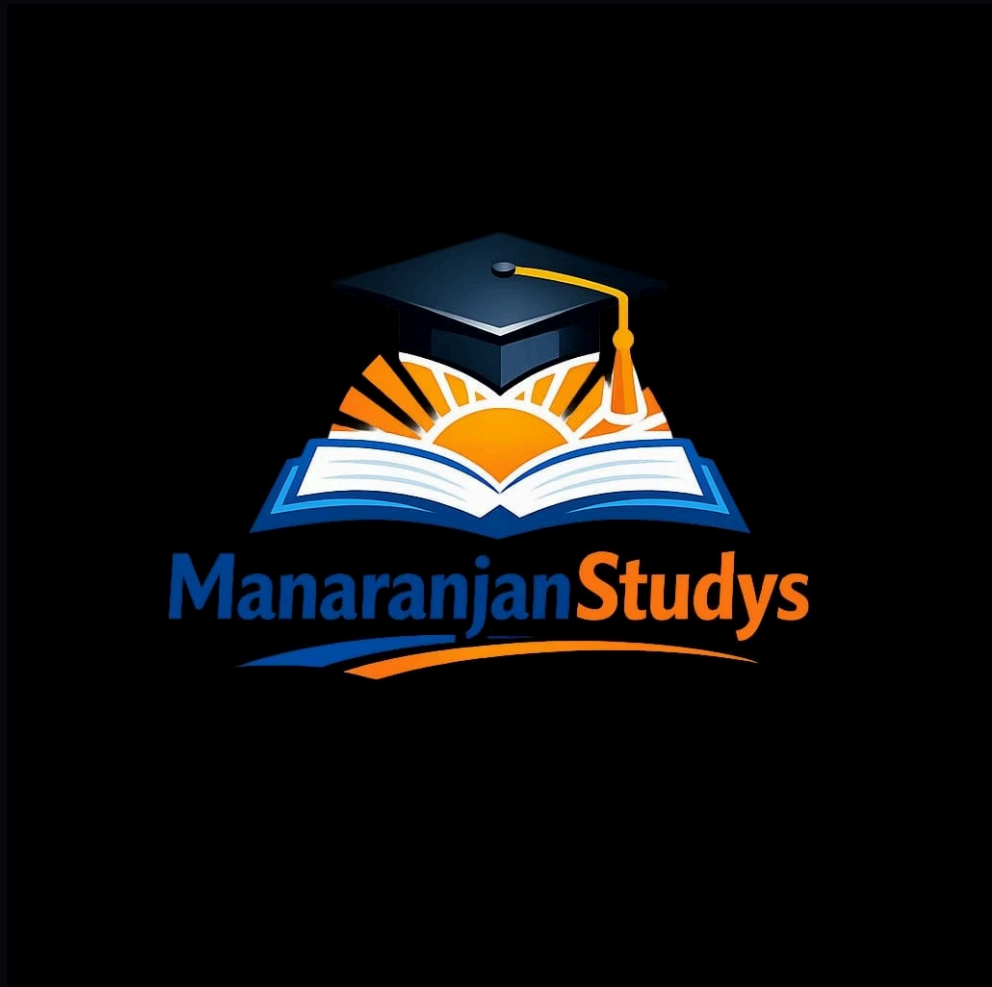
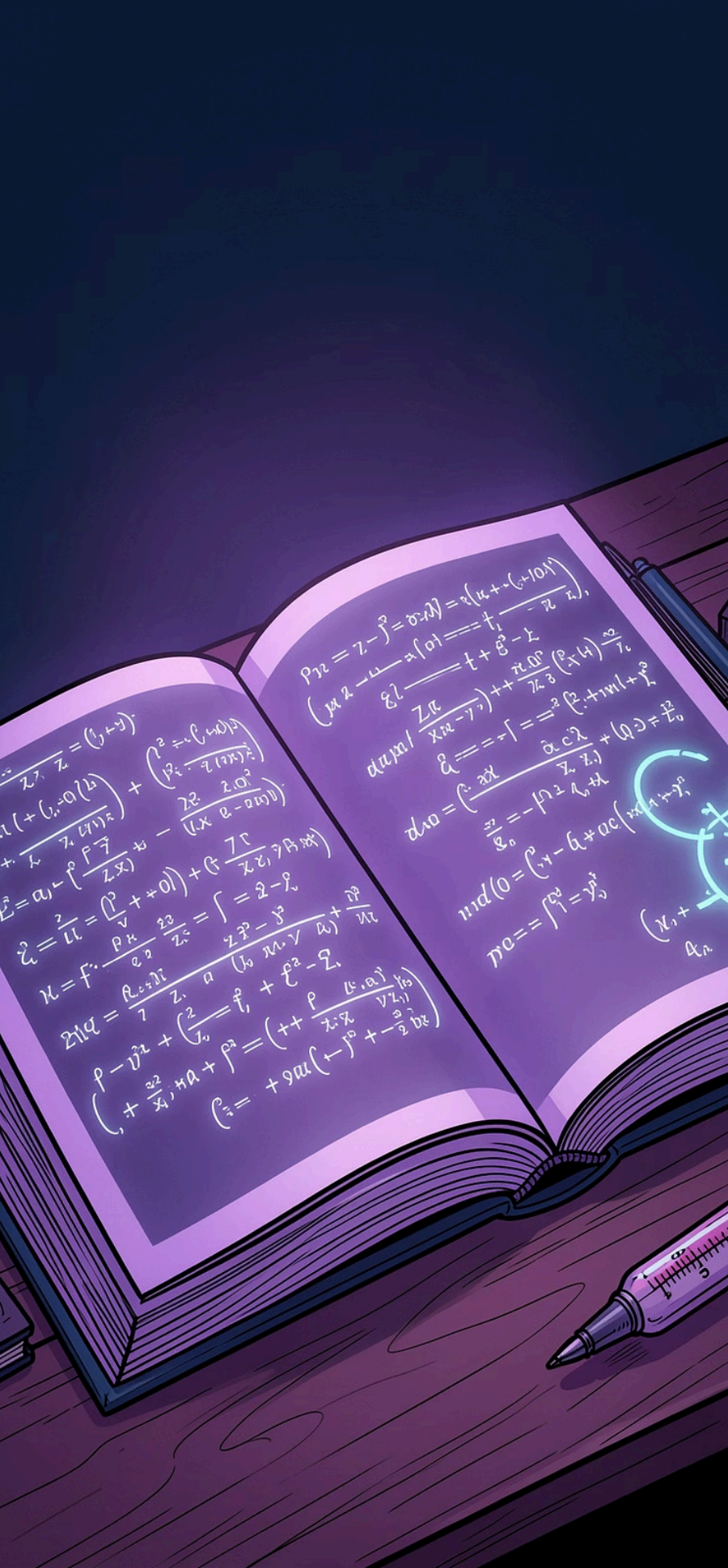




Manaranjan Study's

Chapter 2: Units & Measurement

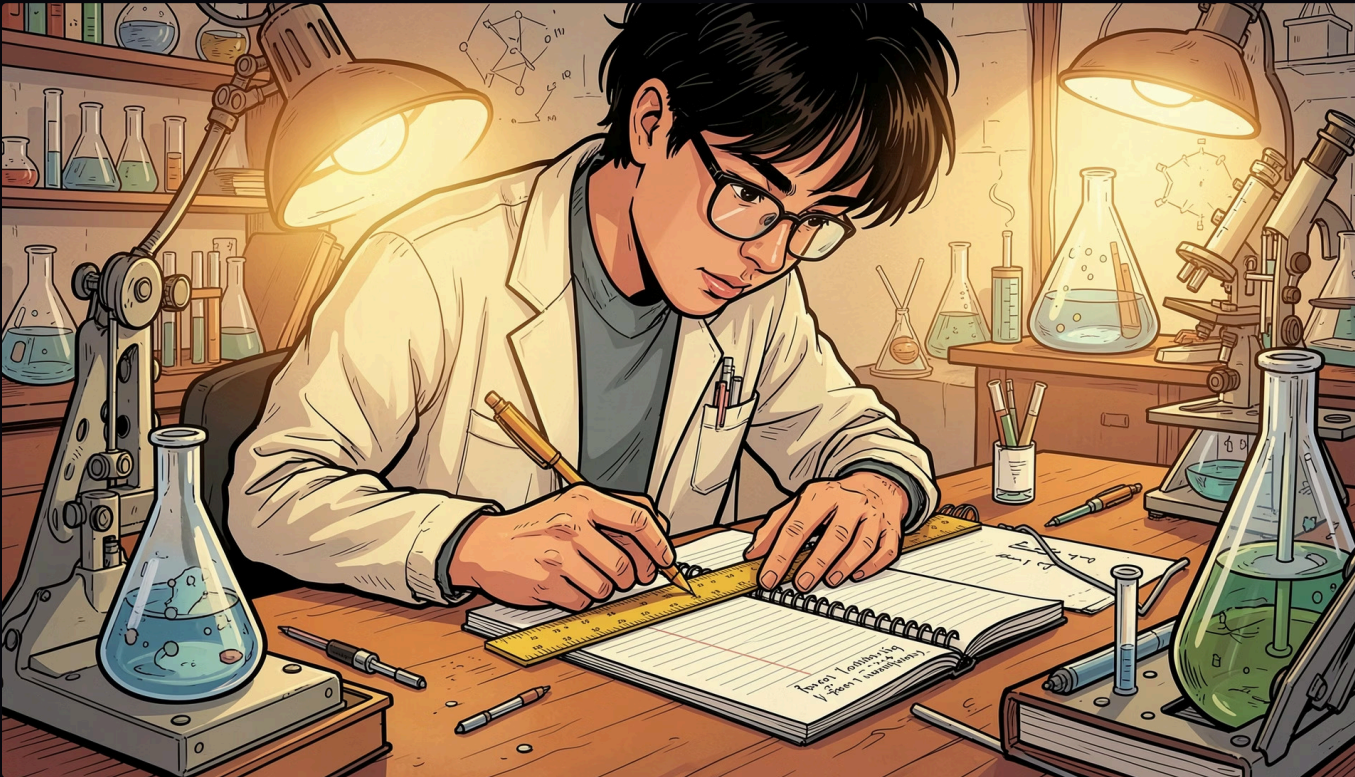
Class 11 Physics • Complete Notes • 5–7 Page Summary

NCERT BASED

CLASS 11 PHYSICS

Made with GAMMA

What is a Physical Quantity?



A **physical quantity** is any property of a physical system that can be measured and expressed numerically with a unit.

Fundamental Quantities

Independent quantities — e.g., Length, Mass, Time

Derived Quantities

Obtained from fundamental ones — e.g., Velocity, Force

Systems of Units

A **system of units** is a complete set of units for all physical quantities. The most widely used system is **SI (Système International)**.

CGS

Centimetre, Gram, Second

MKS

Metre, Kilogram, Second

SI

7 Base Units — International Standard

FPS

Foot, Pound, Second

SI Base Units — The 7 Fundamentals

Quantity	Unit	Symbol	Definition (Brief)
Length	Metre	m	Distance light travels in $1/299792458$ s
Mass	Kilogram	kg	Defined by Planck's constant
Time	Second	s	9,192,631,770 cycles of Cs-133 radiation
Electric Current	Ampere	A	Flow of electric charge
Temperature	Kelvin	K	Thermodynamic temperature scale
Amount of Substance	Mole	mol	6.022×10^{23} particles
Luminous Intensity	Candela	cd	Light intensity in a given direction

Dimensions & Dimensional Formula

Dimensions express a physical quantity in terms of fundamental quantities raised to certain powers.

Example: Force = Mass $\times$ Acceleration

$$[F] = [M][LT^{-2}] = [MLT^{-2}]$$

Velocity

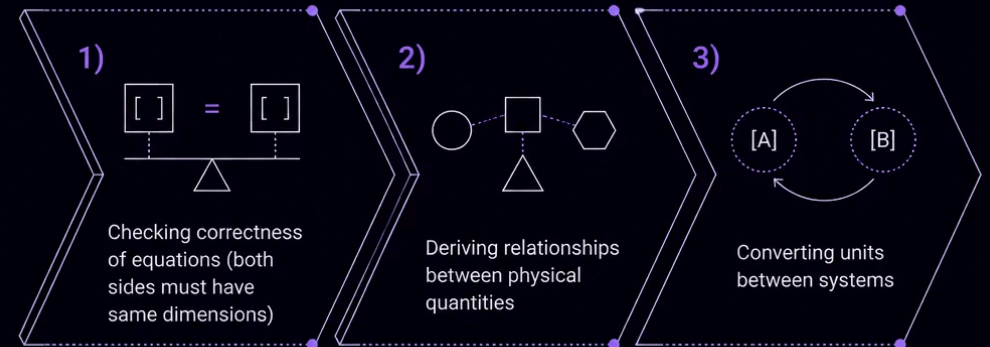
$$[LT^{-1}]$$

Energy

$$[ML^2T^{-2}]$$

Pressure

$$[ML^{-1}T^{-2}]$$



Errors in Measurement

No measurement is perfectly accurate. **Errors** are the difference between measured value and true value.



Systematic Error

Consistent, repeatable error due to faulty instrument or method. Can be corrected.



Random Error

Unpredictable fluctuations. Reduced by taking multiple readings and averaging.



Least Count Error

Smallest value an instrument can measure. E.g., Vernier caliper = 0.01 cm.

Significant Figures



Significant figures are the meaningful digits in a measured value, indicating its precision.

01

All non-zero digits are significant

Example: 234 → 3 significant figures

02

Zeros between non-zero digits are significant

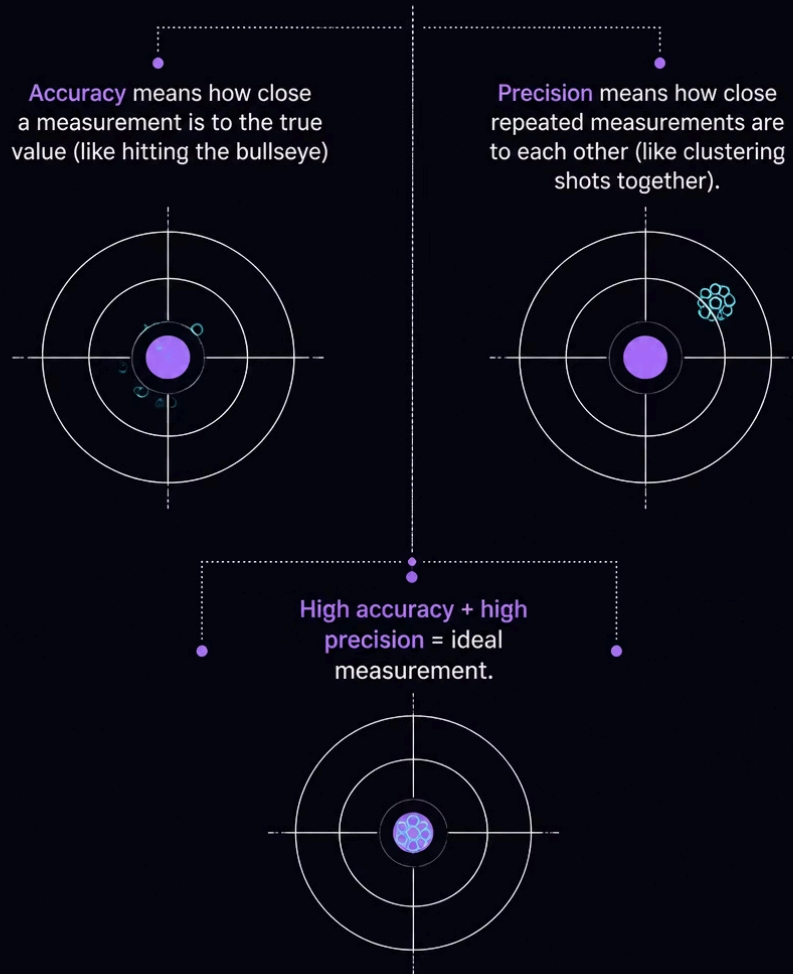
Example: 2007 → 4 significant figures

03

Leading zeros are NOT significant

Example: 0.0045 → 2 significant figures

Accuracy vs Precision



Key Distinction

Accuracy = Closeness to the *true value*

Precision = Closeness of *repeated measurements* to each other

❏ A measurement can be precise but not accurate, and vice versa. The goal is both!

Important Formulas at a Glance



Percentage Error

$$\% \text{ Error} = \frac{|\text{Measured} - \text{True}|}{\text{True}} \times 100$$



Mean Absolute Error

$$\Delta a_{\text{mean}} = \frac{|\Delta a_1| + |\Delta a_2| + \dots + |\Delta a_n|}{n}$$



Dimensional Homogeneity

Both sides of a physical equation must have the **same dimensions**.



Unit Conversion

Multiply by conversion factor: **1 km = 1000 m, 1 hour = 3600 s**

Chapter Summary



Physical Quantities

Fundamental & Derived



SI Units

7 Base Units



Dimensions

[MLT] Formula

notes

Systematic, Random, Least Count



Sig. Figures

Precision in measurement

✔ **Manaranjan Study's** — Revise these concepts regularly for strong fundamentals in Physics. Practice numerical problems on errors and significant figures for exam success!